For our gantry, what would you recommend we use for modeling/simulation before building the physical system?

For position awareness, what would you recommend? (encoders, limit switches, sensors, a combination?)

What payload/weight information do we actually need the software to know? Do we need a sensor that detects load, or is it reasonable for the user to provide the payload weight and have the system reject anything over its limit?

What's a realistic load capacity and frame size for our prototype? What is a realistic prototype size we can expect to make?

Should the frame be ceiling-mounted or freestanding or a combination of both , and how do we make it expandable without losing strength?

Which drive system fits our load and span (lead screw, rack and pinion, or belt), and should the long axis use one motor or two?

What should happen mechanically if power is lost mid-lift? How we should safely handle a stop?

What are the most likely mechanical failures, and which of them can sensors actually detect?